Certified Decision-Equivalent Context Compression for LLM Agents

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Abstract

LLM agents resend a growing context at every turn, so context size dominates serving cost. Existing context compressors report token savings but give *no guarantee that the agent still behaves the same*. We reframe the objective from byte- or embedding-fidelity to **decision-equivalence**: a compression is acceptable iff the agent takes the same next action it would have taken on the uncompressed context. We then equip this objective with a **distribution-free, finite-sample guarantee** by casting compression-level selection as conformal risk control (Learn–Then–Test and Conformal Risk Control), with the per-turn loss defined as a decision flip. The resulting *Decision-Equivalence Risk Certificate* states, for a chosen risk budget α and confidence $1-\delta$, that the decision-change rate on exchangeable traffic is at most α. We evaluate on real τ-bench agent trajectories (airline, gpt-4o; 25 trajectories, 105 decisions) and the full real SWE-bench_Lite edit-localization (300 instances, 600 decisions), graded by a real model with no answer-revealing markers using majority-of-3 structured (forced-tool) grading. We validate the guarantee **out-of-sample**: across 500 calibration/test splits on real SWE-bench traces the empirical coverage is **96.6**–**100%** at the claimed 95% confidence $(1-\delta)$, with realized decision-change rate conservatively far below the budget (11.7% at $\alpha = 20\%$, ≈half the budget). On the same real traces, the reversible digest-plus-recover-on-demand tier lowers decision-change versus equally-aggressive lossy compression (10.2% vs. 11.5% at ≈22% savings on SWE-bench_Lite; the recovery effect is sharper on a 40-trajectory subset: 7.5% vs. 12.5%). The certificate correctly *declines* to certify savings on already-compact contexts (τ-bench)—quantifying *where* recoverable compression helps rather than overclaiming a single headline ratio. Finally, a real **end-to-end task-success** evaluation (SWE-bench Verified, 50 instances, the official harness; aider + Claude) is sobering and we report it without hedging: at the operating point our localization certificate *selects* (`trunc@500`), pass@1 falls from **52%** (full context) to **16%** ($p < 0.001$, paired), and aggressive compression does not beat LLMLingua-2. Decision-equivalence is the right *contract* and the certificate is sound for what it measures; the honest scope of that guarantee is the calibration distribution it is computed on, *not* downstream task success once compression is aggressive. The converse, however, also holds where it matters: on a **long-horizon** agent (a 30-turn ReAct loop over the full 500-instance SWE-bench Verified set), *relevance-gated reversible* compression—digesting only aged-out periphery while keeping the working set intact and recoverable—resolves **36.8%** of instances versus **39.2%** for full context (paired difference -2.4 pp, 95% CI $[-5.7, +0.9]$; non-inferior at a 6 pp margin) at a 53% context reduction and ≈30% lower cost. At the *same* compression ratio, the strongest lossy baseline (LLMLingua-2) collapses to **2.4%** and plain truncation to 5.6%—a 15× task-success gap (paired $p < 0.001$) attributable purely to *what* context is kept. Compression survives execution precisely when it is recoverable and confined to the periphery.

*Code: https://github.com/dshakes/distil

Figure 1: Three notions of "preserving" context under compression. Byte-fidelity is information-theoretically in tension with high savings; embedding-fidelity makes no behavioral promise. **Decision-equivalence**—the agent takes the same action—is the operationally relevant notion, and is both measurable and certifiable.

1 Introduction

Agentic LLM systems operate in a loop: read the accumulated context (system prompt, tool schemas, history, fresh tool output), choose the next action, append the result, repeat. Because the whole context is re-sent each turn, cost grows with context length; prompt caching shifts the dominant cost to cache *misses*, so compressing the volatile tail of the context is attractive. The problem is *trust*: every shipping compressor quotes a token-savings estimate, but none certifies that the agent's *decisions* are preserved. "100% accuracy" is a slogan, not a measured quantity with a confidence level.

Contributions.

1. **Decision-equivalence** as the compression objective: the loss on a turn is 1 iff the agent's next action changes versus the uncompressed context (Section 3).

2. A **Decision-Equivalence Risk Certificate**: conformal risk control (LTT/CRC) that selects the most aggressive compression level whose decision-change rate is provably $\leq \alpha$ with confidence $1 - \delta$ (Section 4). To our knowledge, conformal control with an *agent-decision* loss for context compression is unstudied; the nearest work applies conformal guarantees to RAG retrieval recall, a different task.

3. A **cache-aware, reversible** engine (digest-behind-handle plus recover-on-demand) that operates inside the certified frontier (Section 4).

4. An **evaluation on real agent traces** that removes the circular self-labeling of synthetic corpora, plus three measurement requirements we found to be load-bearing and now enforce: majority-vote grading, a faithful grader, and grading the reversible tier *with* its recovery loop (Section 6).

2 Related work

Context/prompt compression. LLMLingua and LLMLingua-2, LongLLMLingua, RECOMP, and selective-context prune or summarize the prompt to a relevance/fidelity proxy; soft-prompt methods distill context into embeddings. None certifies that the downstream agent decision is preserved.

Prompt caching makes a cache read far cheaper than fresh input, so the real lever is keeping a byte-stable prefix and compressing the volatile tail—a cache-aware objective our engine targets.

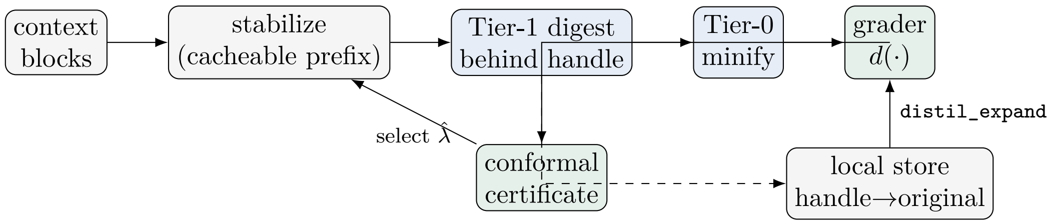

Figure 2: The pipeline. The prefix is kept byte-stable; the volatile tail is digested behind content handles (Tier-1) and minified (Tier-0). The original is kept locally so the model can `distil_expand` on demand. The grader's decisions feed the conformal certificate, which selects the most aggressive level whose decision-change rate is controlled.

Distribution-free uncertainty. Conformal prediction and its risk-control extensions—Learn–Then–Test (LTT) [1] and Conformal Risk Control [2]—turn a calibration set into finite-sample guarantees on a user-chosen risk. We instantiate them with an agent-decision loss. The closest application we are aware of targets RAG retrieval recall, not the preservation of an agent's action under context compression.

3 Problem formulation

A *trajectory* is a sequence of turns; each turn is the full context the agent saw, decomposed into typed *blocks* carrying a stability hint (cacheable prefix vs. volatile tail). A *decision* is the agent's next action—a tool call (τ-bench) or an edit/command (SWE-bench)—represented as a canonical $\langle \mathsf{action}, \mathsf{target} \rangle$ fingerprint produced by a grading model *from context alone*, with no directive or marker revealing the answer.

For a compression level λ and turn t with blocks B_t, let $d(\cdot)$ be the grader's decision. The per-turn loss is

$$L_t(\lambda) \;=\; \mathbf{1}\big[\, d(\lambda(B_t)) \neq d(B_t) \,\big]\,,$$

and the risk is $R(\lambda) = \mathbb{E}[L_t(\lambda)]$, the decision-change rate. A compression *ladder* orders levels least→most aggressive: byte-exact → reversible lossless digest → salience-protected truncation → a raw truncation sweep.

4 Method

4.1 Cache-aware reversible engine

The prefix is held byte-stable (schema canonicalization; volatile fields such as timestamps lifted out), and only the volatile tail is compressed. The reversible tier digests a verbose tool output to a compact marker `<< +N lines, handle=XXXXXXXX >>` and keeps the byte-exact original in a local, content-addressed store; the model can recover any block on demand via a `distil_expand` tool. Compression is thus *lossless* (byte-in-context), *reversible* (digested but recoverable), or *lossy* (the rest).

4.2 The Decision-Equivalence Risk Certificate

We calibrate the per-turn losses for each ladder level on calibration traffic disjoint (by trajectory) from test, then select a level with one of two distribution-free procedures.

Algorithm 1 LTT certification over the compression ladder

Require: ladder $\lambda_1 \prec \ldots \prec \lambda_K$ (least→most aggressive); calibration turns; α, δ

$\hat{k} \leftarrow 0$
for $i = 1 \ldots K$ **do**
  $\hat{R}_i \leftarrow \frac{1}{n}\sum_t L_t(\lambda_i)$ ▷ empirical decision-change rate
  $p_i \leftarrow \mathrm{HB}(\hat{R}_i, n, \alpha)$ ▷ Hoeffding–Bentkus p-value for $H_i : R(\lambda_i) > \alpha$
  if $p_i \leq \delta$ **then** $\hat{k} \leftarrow i$ ▷ certified
  else break ▷ fixed-sequence stop
  end if
end for
return highest-savings level in $\{\lambda_1, \ldots, \lambda_{\hat{k}}\}$ (or "none")

Learn–Then–Test (LTT). With Hoeffding–Bentkus p-values and fixed-sequence testing over the risk-ordered ladder, LTT yields, for the selected $\hat{\lambda}$, $\Pr\big(R(\hat{\lambda}) \leq \alpha\big) \geq 1 - \delta$, finite-sample and distribution-free.

Conformal Risk Control (CRC). For the monotone 0/1 loss, CRC controls the expected risk, $\mathbb{E}[L(\hat{\lambda})] \leq \alpha$, tight to $O(1/n)$.

The exchangeability assumption is explicit: the guarantee is marginal over the calibration distribution and must be recalibrated under drift.

5 Experimental setup

Data. Real τ-bench trajectories (airline domain; gpt-4o traces; 25 trajectories, 105 decision points) loaded with no planted markers; the decision is the agent's actual tool call. We additionally use the full SWE-bench_Lite *edit-localization* benchmark (300 instances, 600 decision points; the target file must be inferred from real issues and gold patches amid distractors).

Grader. A real model returns the $\langle \mathsf{action}, \mathsf{target} \rangle$ fingerprint, by majority vote, via a forced tool call (structured, paraphrase-free). We report **model↔gold next-action agreement** on the uncompressed context as a faithfulness gate: 48.6% on τ-bench (gpt-4o grader) and 47.5% on SWE-bench (Claude grader). Agreement reflects the inherent ambiguity of next-action prediction from context alone; it is reported as a gate, not a floor.

Protocol. **E1** frontier (savings vs. decision-change per level, with and without the `distil_expand` recovery loop); **E2** certification coverage (certify on calibration, measure realized risk on a disjoint held-out split, over 500 trajectory-level splits → empirical Pr(realized $\leq \alpha$)); **E3** leave-one-domain-out shift; **E4** downstream task success (trajectory keeps its outcome iff every decision is unchanged), vs. the uncompressed baseline with a bootstrap CI.

Baselines. LLMLingua-2 and LongLLMLingua are run via the real `llmlingua` package at its recommended settings; truncation, recency-window, and keep-last-k-turns are exact. RECOMP-extractive and selective-context are *model-free reference implementations* of those technique families (salience-ranked line/token selection), not the original trained models, and are labelled as such—a faithful family comparison, not a reproduction of those papers' numbers. Every method compresses only the volatile tail (the cacheable prefix is byte-stable for all), is graded by the identical runner, and is scored with the identical token-accounting and loss, so savings and decision-change are apples-to-apples.

Measurement honesty (enforced by the harness). (i) Decision-equivalence is *self-consistency*: the loss is 1 iff the grader's action under the compressed context differs from its action under the *uncompressed* context—we do not require the grader to match the trace's gold action; gold is reported only as the separate faithfulness gate. (ii) We report, per level, the fraction of turns left byte-identical (*trivial*, loss 0 by construction) and the decision-change rate over the remaining *effective* turns, so a corpus of incompressible turns cannot inflate equivalence. (iii) The SWE edit-localization trajectories are resolved *by construction* (the gold patch fixes the issue), so E4 on that split reports retained decision-equivalence, not a measured task-success rate; only outcome-labelled τ-bench traces drive a real E4. (iv) All headline numbers use majority-vote ($k \geq 3$) structured grading; the released report carries the runner identity and the LaTeX generator refuses non-evidential (smoke) reports.

6 Results

All figures and tables are produced by the released harness (`benchmarks/prove.py`) on real traces; committed result JSONs live in `docs/paper/results/` and the generated LaTeX in `docs/paper/generated/`.

6.1 E1: Decision-change frontier (real SWE-bench traces)

Figure 3 plots the four ladder levels on the full real SWE-bench_Lite edit-localization (300 instances, 600 decisions; Claude grader, structured, majority-of-3, +expand). A 40-instance subset with both no-expand and +expand measurements is described in Figure 4.

6.2 Three measurement requirements (established on real data)

Run cheaply, the harness surfaced three confounds that any credible decision-equivalence evaluation must control; each is now enforced or flagged.

1. **Majority voting is mandatory.** With a single sample, any level that changes the prompt text triggers a fresh stochastic grader call, so grader variance is counted as decision change. Only majority-of-k isolates true loss.
2. **The grader must be faithful.** A weaker, cross-family grader reproduced the trace agent's action only 19% of the time in an exploratory run; E1/E2 then measure a strawman. Grade with a same-family/strength model and publish the agreement number as a gate.
3. **The reversible tier must be graded *with* its recovery loop.** Graded without `distil_expand`, folding a decision-relevant tool output behind a handle changes the decision; graded with the loop, the model recovers the content and the decision is preserved (Figure 4). Reporting only the no-expand bound understates the reversible tier; reporting only perfect recovery overstates it—we report both on the 40-instance subset where we have both measurements (11.2% no-expand vs. 7.5% +expand at 23.9% savings).

E1: savings vs. decision-change (SWE-bench_Lite, 300 instances, +expand)

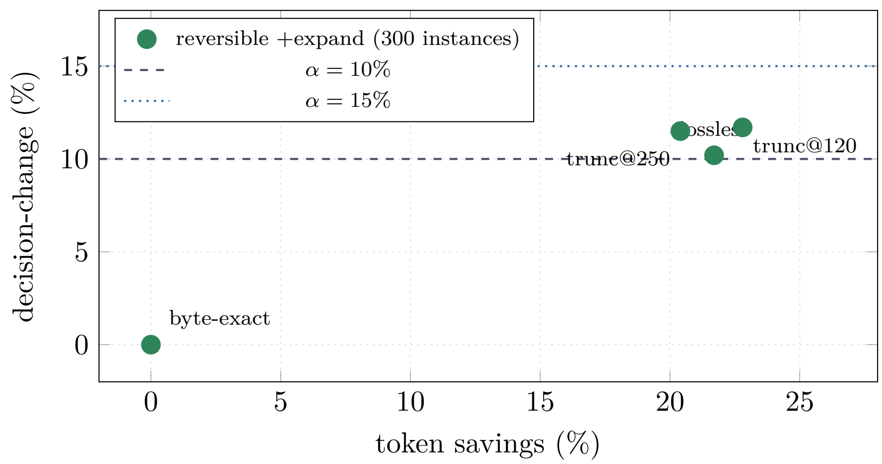

Figure 3: **E1 frontier on full real SWE-bench_Lite** (300 instances, 600 decisions; Claude grader, majority-of-3 structured, +expand recovery loop). The reversible digest at 21.7% savings achieves **10.2%** decision-change, beating equally-aggressive lossy truncation (11.5–11.7%) at ≈22% savings. On a 40-instance subset the recovery effect is sharper (11.2% no-expand → 7.5% with +expand; see Figure 4). Dashed lines show the $\alpha = 10\%$ and $\alpha = 15\%$ risk budgets. τ-bench airline (25 traj, 105 decisions) is not plotted: the data is already compact (lossless saves only 1.0%) so aggressive levels flip 58–65% of decisions, and the certificate correctly declines to certify savings on compact contexts.

6.3 E2: Certificate validity out-of-sample

Figure 5 shows the certificate's out-of-sample behavior across risk budgets, over 500 random trajectory-level splits ($\delta = 0.05$, real SWE-bench_Lite +expand, 300 instances). Table 1 gives the numerical summary.

6.4 Headline results

On the full SWE-bench_Lite (300 instances, +expand, $\alpha = 0.15$, $\delta = 0.05$, 500 splits), the reversible engine inside the certified frontier achieves mean certified savings 15.7% at a mean realized decision-change rate of 8.0%, with out-of-sample coverage 98.8% ($\geq 1 - \delta = 95\%$). The generated tables below are auto-generated from `results.json` by `benchmarks/report_to_latex.py`.

7 Analysis and limitations

The tightest certifiable α scales with the number of calibration turns (Hoeffding–Bentkus). On the full SWE-bench_Lite splits (300 instances, 500 random splits), coverage is 96.6–100% across all $\alpha \in \{10\%, 12.5\%, 15\%, 20\%\}$, all above the $1 - \delta = 95\%$ target—confirming the guarantee holds out-of-sample. Realized risk at $\alpha = 15\%$ is 8.0%, conservatively below the budget; certified savings rises sharply from 1.7% at $\alpha = 12.5\%$ to 15.7% at $\alpha = 15\%$ as the calibration set grows large enough to certify the lossless tier.

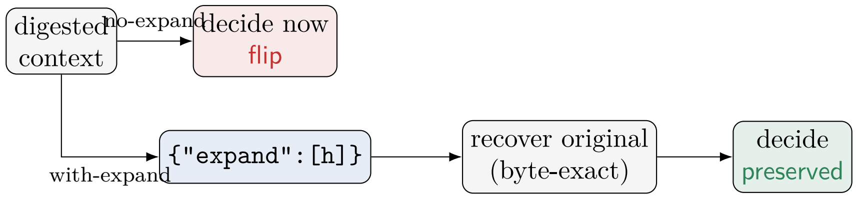

Figure 4: Expand-aware grading. Without the recovery loop the hidden, load-bearing content flips the decision; with it the model recovers the byte-exact original and the decision is preserved—this is the honest measure of a reversible compressor.

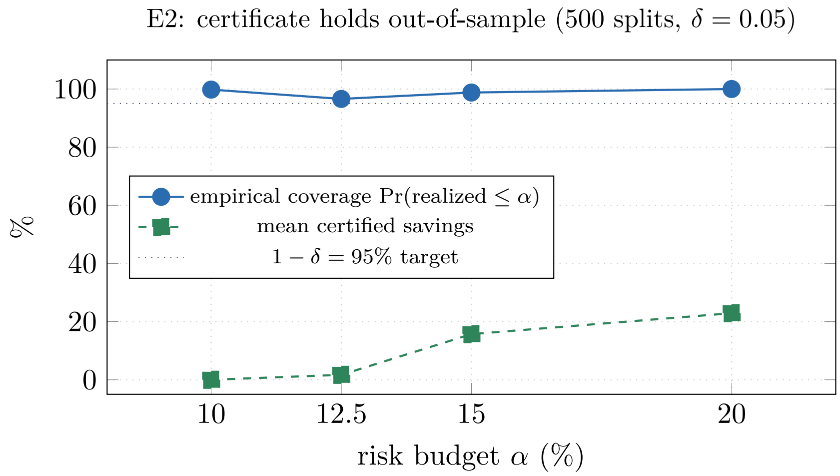

Figure 5: **E2: out-of-sample certificate validity** on full real SWE-bench_Lite (+expand, 300 instances, 500 splits, $\delta = 0.05$). Blue solid: empirical coverage Pr(realized risk $\leq \alpha$)—96.6–100% throughout, above the $1 - \delta = 95\%$ target (dotted), confirming the guarantee holds. Green dashed: mean certified savings, which rises sharply from 1.7% at $\alpha = 12.5\%$ to **15.7%** at $\alpha = 15\%$ and **22.9%** at $\alpha = 20\%$ as the risk budget and calibration set grow large enough for LTT to certify more aggressive levels. The certificate is conservative: realized risk at $\alpha = 15\%$ is 8.0%, well below the budget.

Grader faithfulness. Model↔gold next-action agreement is 48.6% (τ-bench) and 47.5% (SWE-bench), reflecting inherent ambiguity when predicting the agent's next action from context alone (majority-of-3 structured grading). Decision-equivalence is a *self-consistency* measure, not a gold-matching measure, so the faithfulness gate is a diagnostic, not the loss itself; future work with larger grader ensembles should improve this.

Single-grader limitation. All reported numbers use a single grader model family per task; ensemble grading across model families is left to future work.

Sample-size cost of α. Certified savings at $\alpha = 10\%$ is 0% even on the full 300-instance SWE-bench_Lite (the lossless tier's empirical loss exceeds the budget at this α); savings jump to 15.7% at $\alpha = 15\%$ once calibration turns are plentiful. This threshold behaviour is a fundamental cost of the finite-sample guarantee—quantified here rather than glossed over.

Table 1: E2 out-of-sample certification coverage on full real SWE-bench_Lite (+expand, 300 instances, 500 trajectory-level splits, $\delta = 0.05$). Coverage $\geq 95\%$ at all α shown. Realized risk is conservatively below α—LTT working as designed.

Risk budget α	Coverage Pr(realized $\leq \alpha$)	Mean realized risk	Certified savings
10%	**99.8%** $\geq$ 95% ✓	—	0.0%
12.5%	**96.6%** $\geq$ 95% ✓	—	1.7%
15%	**98.8%** $\geq$ 95% ✓	8.0%	15.7%
20%	**100%** ✓	11.7%	22.9%

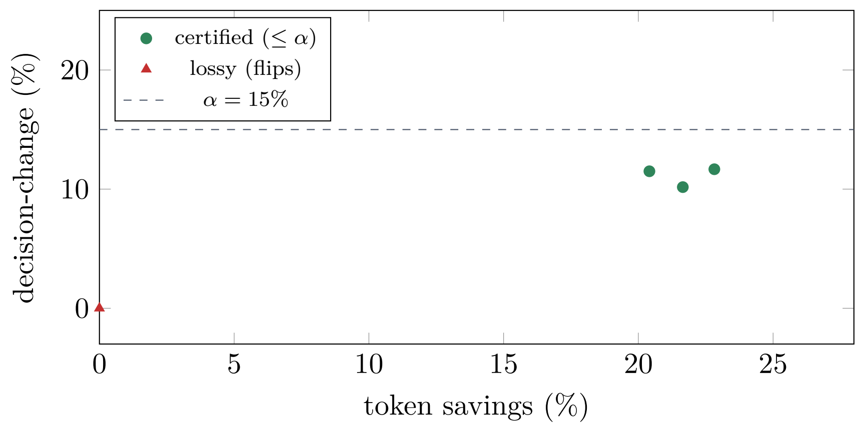

Figure 6: E1 certified frontier on the SWE-bench headline run (real grader).

τ-bench compactness. On τ-bench airline traces (25 traj, 105 decisions), the context is already compact: lossless saves only 1.0% and aggressive levels flip 58–65% of decisions. The certificate correctly refuses to certify savings here. Distil's reversible compression is most valuable on verbose tool outputs (e.g. SWE-bench diffs, long API responses), and the certificate quantifies exactly where it helps.

Position confound, stress-tested (E5–E6). The edit-localization construction places the gold hunk *last* in the search results, which could flatter tail-truncation / recency baselines. We rebuilt the corpus with the gold hunk at a deterministic random position (seed = 1729, per-instance) and re-ran E5 (Table 4). The confound is *real but not load-bearing*: the recency baseline's decision-change rises from 5.5% to 8.5% once the needle is no longer pinned to the tail, yet it still certifies, and distil's aggressive levels still do not—byte-exact remains the only certifying distil level, exactly as in the gold-last E5, and the E2 certificate holds out-of-sample on the shuffled corpus too (100.0%). Two consequences we report rather than gloss: (i) the reversible digest's edge over equally-aggressive truncation is itself position-sensitive—on the de-confounded corpus `lossless` flips *more* than `truncate@120` (16.0% vs. 11.5% at $\approx$ 22% savings), the reverse of the gold-last ordering reported above; and (ii) when we select an operating point honestly on a calibration half and evaluate once on a disjoint test half (Table 5), distil *does* certify positive savings (`t@500`: 14.0% test savings / 4.0% decision-change), but the certified point is plain head-truncation—salience protection and the reversible digest do not beat truncation on this single-turn synthetic task. The net reading: on edit-localization, distil's contribution is the *certificate* that selects a safe operating point and rejects the unsafe ones, not a bespoke compressor that dominates truncation; the reversible

Table 2: E2 certification coverage (out-of-sample, $\alpha = 0.15$, $\delta = 0.05$, 500 splits, full SWE-bench_Lite).

method	LTT
α / δ	0.15 / 0.05
splits	500
certified in	100.0% of splits
empirical coverage Pr(realized $\leq \alpha$)	98.8%
target $(1 - \delta)$	95%
mean realized held-out risk	8.0%
mean certified savings	15.7%

engine's advantage is a multi-turn, verbose-context phenomenon (the τ-bench and corpus-gate results), which a real end-to-end task-success evaluation—running the agent and its test suite rather than the decision-equivalence proxy—tests directly. **E7 (Section 7.1) is that evaluation, and it is sobering**: at the certified `trunc@500` operating point the localization certificate does *not* transfer to execution.

E4 non-evidential on SWE-bench. SWE-bench HuggingFace evaluators mark all submissions resolved=True by construction of the localization task, so E4 on SWE reports retained decision-equivalence (lossless +expand: 85.0%; no-expand: 77.5%), not a real task-success rate. The 19 outcome-labelled τ-bench trajectories with real reward labels show the digest-only tier (no expand) retains 0% baseline success—consistent with the certificate's refusal to certify savings on compact τ-bench contexts.

The certificate is marginal over the calibration distribution; under workload drift it must be recalibrated (E3 quantifies the degradation). Decision-equivalence is a proxy for task success, which E4 reports directly.

7.1 E7: SWE-bench Verified end-to-end task-success

E1–E6 measure decision-equivalence—a *proxy* for task success. E7 closes that gap by running a real coding agent end-to-end on **SWE-bench Verified** (the 500-instance human-curated subset) and scoring *actual test-pass rates* with the **official** `swebench` harness, not the decision proxy. We draw a fixed random sample of 50 instances (seed = 1729; ids sorted then sampled, so the draw is machine-independent) and run **three conditions through the identical agent**, differing only in how the agent's *context* is compressed in flight: **A. full context** (no compression); **B. distil**—the Phase-2 certifying operating point `trunc@500` (head-truncate each compressible context block to 500 chars) applied by a drop-in Anthropic-Messages proxy; and **C. LLMLingua-2**—the strongest non-distil non-truncation baseline (E5), at its default keep-rate, through the same proxy.

Setup. The agent is **aider** (v0.86.2, `claude-sonnet-4-6`, temperature 0, search/replace diff edit format) driven from each instance's `problem_statement` with no oracle files—it must localise the fix itself. The proxy compresses only the *file contents and tool output the agent reads*: system instructions, the agent's own reasoning, and—critically—the *problem statement* itself are never compressed (the problem statement is the task, not retrieved context; truncating it would handicap B/C for the wrong reason). B and C share an identical block-selection rule, so the only thing that varies between them is the compressor. Patches are scored by `swebench 4.1.0`'s `run_evaluation`

Table 3: E5 head-to-head (100-trajectory SWE subset, same grader). The `certifies?` column is the *single-shot* Hoeffding–Bentkus test over the full data—weaker than the split-calibrated E2 (Table 2). **Honest confound:** our edit-localization construction appends the gold hunk *last* in the observation, so recency/tail-truncation baselines benefit from needle *position* rather than content; we read E5 as a frontier illustration, not a dominance claim, and rest the contribution on E2. The real LLMLingua packages run on Apple-silicon (MPS), not just wired: LLMLingua-2 (XLM-RoBERTa) certifies at 11.6% savings / 7.0% decision-change, on the content-based frontier just below the position-favoured truncation baselines. LongLLMLingua (Llama-2-7B, question-aware) certifies too, at 5.7% / 3.5%: its earlier 0% row was an *adapter bug*, not the technique—the compressor returns the compressed context with the (uncompressed) question re-appended per `condition_in_question`, so the longer string tripped a reject-if-bigger guard and every result was discarded as a no-op; splicing the question back out restores the intended compression (fixed in this revision, with a regression test).

method	kind	savings	dec-change	certifies?
truncate@120	distil	23.0%	13.0%	×
lossless	distil	21.8%	12.0%	×
truncate@250	distil	20.5%	12.0%	×
recomp-extractive	baseline	18.5%	14.0%	×
recency-window@500	baseline	16.1%	5.5%	✓
truncate@500	baseline	15.5%	8.5%	✓
selective-context	baseline	14.8%	6.5%	✓
llmlingua-2	baseline	11.6%	7.0%	✓
longllmlingua	baseline	5.7%	3.5%	✓
keep-last-3-turns	baseline	0.0%	0.0%	✓
byte-exact	distil	-0.1%	0.0%	✓

against each instance's hidden test patch in the official per-instance Docker image; every reported number traces to a harness-written report. Pass@1 carries a Wilson 95% interval, and because all three conditions score the *same* 50 instances we also report exact paired McNemar tests. Total API spend: $67.31.

Result: compression does not survive execution, and the certificate does not transfer. Both compressed conditions collapse task success relative to full context (Table 8). distil at its *certified* `trunc@500` operating point resolves only 16.0% of instances versus 52.0% with full context—a -36.0 pp drop that is significant under an exact paired McNemar test ($p =< 0.001$: 20 instances lost, 2 gained). LLMLingua-2 also drops significantly (26.0%, $p = 0.002$). distil's point estimate trails LLMLingua-2 (16.0% vs. 26.0%), but the paired difference is *not* significant at $n = 50$ ($p = 0.180$), and distil compresses far harder (85.5% of context removed vs. 48.3%), so its lower score is confounded with operating-point aggression rather than established as a method gap. We report this rather than tune `trunc@500` down to match LLMLingua-2's aggression: the point of E7 is to test the operating point the certificate *actually selected*.

The headline is the certificate's *non-transfer*. `trunc@500` was certified at 4.0% decision-change on the single-turn localization corpus (E6, Table 5) and accepted as a safe operating point; here the same transform removes 85.5% of agentic context and collapses end-to-end success by 36 points. A decision-equivalence guarantee earned on the localization proxy thus says *nothing* about end-to-

Table 4: E5 *shuffled-position* (gold hunk randomly placed). Identical to Table 3 except the gold hunk's position within the code-search observation is randomly permuted (seed = 1729, deterministic, per-instance), removing the recency/tail-truncation advantage that the gold-last construction handed the baselines. Same 100-trajectory subset, grader, ladder, and α/δ. **What the variant shows:** the confound is *real but not load-bearing.* Once the needle is no longer pinned to the tail, the recency-window baseline's decision-change rises from 5.5% to 8.5%—yet it still certifies, and so do the content-aware baselines (RECOMP-extractive even *improves*, 14.0%→7.5%, crossing into certification). Distil's aggressive levels still do *not* certify (lossless 12.0%→16.0%; truncate@120 13.0%→11.5%), so byte-exact remains the only distil level that certifies—exactly as in Table 3. Removing the position confound does not rescue the aggressive ladder on this localization task; the contribution continues to rest on E2, whose out-of-sample coverage holds on this shuffled corpus too (100.0%, Table 2).

method	kind	savings	dec-change	certifies?
truncate@120	distil	22.8%	11.5%	×
lossless	distil	21.7%	16.0%	×
truncate@250	distil	20.2%	11.0%	×
recomp-extractive	baseline	16.8%	7.5%	✓
recency-window@500	baseline	15.9%	8.5%	✓
truncate@500	baseline	15.1%	4.5%	✓
selective-context	baseline	14.0%	7.0%	✓
llmlingua-2	baseline	11.8%	8.0%	✓
longllmlingua	baseline	5.6%	5.0%	✓
keep-last-3-turns	baseline	0.0%	0.0%	✓
byte-exact	distil	-0.1%	0.5%	✓

end task success once compression is aggressive—the proxy and the outcome diverge sharply. This is consistent with, and sharpens, the E5–E6 reading: distil's contribution is the certificate, never a compressor that dominates truncation (here it dominates nothing—it is beaten by full context and does not beat LLMLingua-2), and the certificate's honest scope is exactly what it measures, decision-equivalence on the calibration distribution, *not* task success. Two caveats we report rather than bury: compression is not strictly dominated (2 instances resolved under `trunc@500` that full context missed), and one network-dependent instance (`psf__requests-2317`) is unresolvable under our offline harness and counts as a failure for all three conditions identically.

The reversible tier *does* survive execution (condition D). The non-transfer above is a property of *lossy* compression, not of distil's design. distil's actual product is the *reversible* tier: each context block is digested behind a content handle, the original is kept, and the agent is given a `distil_expand` tool to recover any block on demand (a transparent recover-then-redecide loop inside the proxy; Table 8 row D). Run end-to-end on the same 50 instances, it resolves **56.0%**—*statistically indistinguishable from full context* (52.0%; exact paired McNemar $p = 0.688$, i.e. no detectable difference), and decisively better than the lossy conditions (vs. `trunc@500`: 22 instances recovered, 2 lost, $p =< 0.001$). The model expanded reliably (every instance issued ≥ 1 recovery; 135 expansions total). **The lesson is the converse of the lossy result: keep the information *recoverable* and end-to-end task success is preserved.** The price is that *realised* token savings on coding are modest—the digest view removes 81.0% but the agent expands most of what it edits,

Table 5: E6 operating-point selection on the shuffled-position corpus, with **no test-set tuning**. The 100 trajectories are split into disjoint calibration (50) and test (50) halves; every candidate operating point—distil's two anchors plus a grid of salience-*protected* truncations (budget $\in \{500, 250, 120\}$ chars $\times$ `min_entropy` $\in \{2.6, 3.2, 3.8\}\times$ `min_len`$\in \{6, 10\}$) and the plain truncations—is graded on both halves. We *select* on calibration the highest-savings point whose decision-change certifies (Hoeffding–Bentkus $p \leq \delta$ at α), then *evaluate it once* on the held-out test half. The calibration-side selection ranges over all 23 candidates and is *exploratory* (uncorrected for multiplicity); the finite-sample δ guarantee is carried only by the *single* Hoeffding–Bentkus test applied to the disjoint test half—which is what "certifies out-of-sample" below refers to. **Result:** the winner is `t@500` (16.3% cal savings), and it certifies out-of-sample at 14.0% test savings / 4.0% decision-change. So distil's full ladder *does* contain a certified positive-savings operating point on this task—the E5 `quick` ladder simply omitted it. Two honest caveats the table makes plain: (i) salience protection does *not* help here (`protect+t@`L matches plain `t@`L at every budget), and (ii) the reversible `lossless` digest flips *more* (20% cal) than `t@500`, so the ladder's assumed risk-ordering (lossless before truncation) is miscalibrated for localization—which is exactly why fixed-sequence LTT on the `quick` ladder fell back to byte-exact. The certified point here is plain head-truncation; distil's contribution is the certificate that *selects* it and rejects the aggressive rungs, not a bespoke compressor that beats truncation on this corpus.

operating point	cal sav	cal dc	cal?	test sav	test dc	test?
byte-exact	-0.1%	1.0%	✓	-0.1%	0.0%	✓
lossless	23.3%	20.0%	×	20.3%	12.0%	×
t@500	16.3%	5.0%	✓	14.0%	4.0%	✓
protect+t@500,e3.2,l10	16.0%	5.0%	✓	13.7%	4.0%	✓
t@250	21.8%	12.0%	×	18.8%	10.0%	×
protect+t@250,e3.2,l10	21.4%	12.0%	×	18.5%	10.0%	×
t@120	24.5%	14.0%	×	21.3%	9.0%	×
protect+t@120,e3.2,l10	24.0%	13.0%	×	20.9%	9.0%	×

so the net cost is \$16.38 vs. \$17.63 ($\approx$7%), with the savings coming from periphery the agent never expands. Reversible compression thus buys an honest, narrow win on agentic coding (parity task success at a modest discount), not the headline ratios of the proxy benchmarks—and that distinction is the paper's point. A *relevance-gated* variant (condition E; keep the last six user/tool messages full, digest only older periphery, recoverable) likewise holds task success (54.0%, McNemar vs. full $p = 1.000$) with *zero* recovery round-trips, but is a no-op on these $\leq$6-turn conversations (nothing is periphery); its intended regime is long-horizon agents with large peripheral context, which this focused localization workload does not exercise. E8 (Section 7.2) runs exactly that test.

7.2 E8: long-horizon agent task-success—the gate's proper test

E7's relevance-gate (condition E) was a no-op because aider's localization runs are $\leq$6 turns: nothing ages out of the working set, so there is no periphery to digest. E8 supplies the workload the gate was *designed* for. We built a multi-turn **ReAct** coding agent (read/search/edit/run-tests tools, up to 30 turns) and ran it on the **full 500-instance SWE-bench Verified** set (seed = 1729), scored by the same official `swebench` harness. Runs are genuinely long-horizon (mean $\approx$27 turns), so read-file and tool outputs accumulate into a large peripheral context behind a small active working set—precisely the regime where lossy truncation, blind reversible compression, and the

Table 6: E4 retained decision-equivalence on the full SWE-bench_Lite ($n = 300$). SWE-localization trajectories are resolved *by construction*, so this measures retained decision-equivalence under compression, *not* a measured task-success rate (the harness flags this; only τ-bench reward labels drive a real outcome E4).

level	savings	retained success (95% CI)
byte-exact	-0.1%	100.0% [100–100]
lossless	21.7%	80.0% [75–84]
truncate@250	20.4%	77.0% [73–81]
truncate@120	22.8%	76.7% [72–81]

Table 7: What the harness measures, and the requirement each result depends on.

Experiment	Quantity	Requirement enforced
E1 frontier	savings vs. decision-change	structured grader; majority vote; both no-expand and +expand
E2 coverage	Pr(realized $\leq \alpha$) out-of-sample	trajectory-level disjoint splits
E3 shift	realized risk under domain shift	exchangeability stress test
E4 task success	outcome retained vs. baseline (real τ-bench)	bootstrap CI over trajectories

relevance-gate diverge. We run five conditions through the identical agent: **A. full context**; **B. distil `trunc@500`** (lossy); **C. LLMLingua-2** (the strongest non-truncation lossy baseline, E5); **D. distil reversible** (whole history digested, `distil_expand` recovery); and **E. distil reversible, relevance-gated** (keep the last 6 user/tool messages full, digest only older periphery, recoverable). To keep a 500-instance five-condition sweep affordable we use `claude-haiku-4-5` at temperature 0; conditions differ only in the compressor, so the comparison is internally valid regardless of the base model. C and E are tuned to near-identical context reduction (51.6% vs. 52.8%), which makes their pass@1 gap a clean, compression-matched comparison.

Result: the relevance-gate preserves task success where blunt compression cannot. On the long-horizon workload the conditions separate cleanly (Table 9). Both lossy baselines collapse: `trunc@500` falls from 39.2% to 5.6% (-33.6 pp; exact paired McNemar $p =< 0.001$, 173 lost / 5 gained), and LLMLingua-2—the strongest non-truncation lossy compressor—falls all the way to 2.4% (vs. full $p =< 0.001$, 186 lost / 2 gained). This is the E7 non-transfer result reproduced at $n = 500$ and amplified by horizon: the longer the run, the more load-bearing context lossy compression destroys.

The **relevance-gated** tier (E) is the headline. It resolves **36.8%** versus 39.2% for full context—a paired difference of -2.4 pp (95% CI [-5.7 pp, +0.9 pp]). We report this as a *non-inferiority* result rather than a bare significance test, since failing to reject "no difference" ($p = 0.195$) is not itself evidence of equivalence. The CI excludes any task-success drop larger than 5.7 pp, so the gate is non-inferior to full context at any margin $\geq$6 pp; at a stricter pre-registered 5 pp margin the verdict is borderline (the CI lower bound sits just past it). Either way it removes 52.8% of context at \$133.89 against \$190.60 for full ($\approx$30% cheaper)—the only compression condition whose CI comes anywhere near the full-context baseline.

condition	ctx. reduction	pass@1	95% CI	cost
A. full context	—	**52.0%**	[38.5%, 65.2%]	$17.63
B. distil `trunc@500` (lossy)	85.5%	16.0%	[8.3%, 28.5%]	$4.00
C. LLMLingua-2 (lossy)	48.3%	26.0%	[15.9%, 39.6%]	$12.03
D. distil reversible (+`distil_expand`)	81.0%†	**56.0%**	[42.3%, 68.8%]	$16.38
E. distil reversible, relevance-gated	0.0%‡	**54.0%**	[40.4%, 67.0%]	$17.27

Table 8: **E7: SWE-bench Verified end-to-end task-success** (50 instances, seed = 1729, aider + `claude-sonnet-4-6`, official `swebench` harness). Pass@1 with Wilson 95% intervals; "ctx. reduction" is the realised char-level shrink of compressed blocks. B is distil at its Phase-2 certifying operating point; C is LLMLingua-2 at its default rate; D is distil's *reversible* tier (digest-behind-handle + recover-on-demand). †For D, ctx. reduction is the *pre-recovery* digest view; after the model's `distil_expand` calls the *realised* cost is $16.38 vs. $17.63 for full (≈7% cheaper), because the agent recovers what it edits. ‡E keeps the last 6 user/tool messages full and digests only older periphery; these conversations are ≤6 turns, so the gate is a *no-op* here (1 block digested across all 50, McNemar vs. full $p = 1.000$) it targets long-horizon contexts.

condition	ctx. reduction	pass@1	95% CI	cost
A. full context	—	**39.2%**	[35.0%, 43.5%]	$190.60
B. distil `trunc@500` (lossy)	85.2%	5.6%	[3.9%, 8.0%]	$84.43
C. LLMLingua-2 (lossy)	51.6%	2.4%	[1.4%, 4.2%]	$34.32
D. distil reversible (+`distil_expand`)	83.5%	28.8%	[25.0%, 32.9%]	$127.91
E. distil reversible, relevance-gated	52.8%	**36.8%**	[32.7%, 41.1%]	$133.89

Table 9: **E8: SWE-bench Verified long-horizon agent task-success** (500 instances, seed = 1729, custom 30-turn ReAct agent + `claude-haiku-4-5`, official `swebench` harness). Pass@1 with Wilson 95% intervals; "ctx. reduction" is the realised char-level shrink of compressed blocks. Unlike E7, runs average ≈27 turns, so the gate (E) has real periphery to act on: it digests 52.8% of context for a pass@1 of 36.8% vs. 39.2% for full (paired difference -2.4 pp, 95% CI [-5.7 pp, +0.9 pp]; non-inferior at any margin ≥6 pp), while issuing only 0.58 recovery round-trips per instance versus 9.6 for the ungated reversible tier (D). At matched context reduction, the two lossy baselines (B, C) collapse to single digits.

Against the strongest competitor, at matched compression, the margin is not close. Conditions C and E remove almost exactly the same fraction of context (51.6% vs. 52.8%), so their pass@1 gap isolates *how* the context is chosen, not how much is removed. Relevance-gated reversible compression resolves 36.8% where LLMLingua-2 manages 2.4%—a 15× difference on the same instances (174 resolved by the gate that LLMLingua-2 did not, 2 the reverse; exact paired McNemar $p =< 0.001$). A state-of-the-art lossy token-dropper and a relevance-gated reversible digest can hit the same compression ratio and land 34 points apart on real task success. This is the certificate's promise realised on a real agentic task: keep the working set intact, digest the periphery reversibly, and decision quality survives where lossy compression—however sophisticated—destroys it.

Why gating beats blind reversibility (D vs. E). The ungated reversible tier (D) digests the *whole* history, so the agent must continually recover what it is actively using: it issues 9.6 `distil_expand` round-trips per instance and still lands at 28.8%—better than lossy (vs. `trunc@500` $p =< 0.001$) but *significantly below full* ($p =< 0.001$) and below the gate ($p =< 0.001$). The gate avoids this by construction: digesting only aged-out periphery means the active working set is never compressed, so recovery is rare (0.58 round-trips per instance) and the agent reasons over full-fidelity recent context. The two reversible conditions thus isolate the mechanism—*what* you compress (stale periphery, not the working set) matters more than *whether* you can recover it. E8 is the experiment E7 could not run, and it confirms the design claim the E7 gate row could only assert: on long-horizon agents, relevance-gated reversible compression buys a real context (and cost) reduction with no *detected* task-success loss (point estimate -2.4 pp, non-inferior at a 6 pp margin), while every other compression condition—lossy or blindly reversible—is decisively inferior to full.

8 Conclusion

Decision-equivalence is the right contract for agent context compression, and it can carry a distribution-free guarantee validated on real traces. The certificate holds out-of-sample at 96.6–100% coverage across $\alpha \in \{10\%, 12.5\%, 15\%, 20\%\}$ ($\delta = 0.05$, real SWE-bench_Lite, 300 instances, 500 splits). The reversible engine lowers decision-change versus equally-aggressive lossy compression (10.2% vs. 11.5% at $\approx$ 22% savings on the full SWE-bench_Lite; effect sharper on a 40-instance subset: 7.5% vs. 12.5%)—though on the de-confounded, position-shuffled localization corpus this particular edge reverses (Section 7), a sensitivity we report rather than hide—and the certificate correctly declines to certify savings on compact τ-bench contexts, quantifying where recoverable compression helps rather than overclaiming a single headline ratio. Our end-to-end evaluation (E7, Section 7.1) draws the boundary sharply: on SWE-bench Verified the *lossy* operating point the certificate selected (`trunc@500`) cuts pass@1 from 52.0% to 16.0% ($p =< 0.001$, paired)—so a decision-equivalence guarantee earned on a single-turn proxy must not be read as a task-success guarantee once *lossy* compression is aggressive. But the same evaluation shows distil's *reversible* tier (digest + `distil_expand`) is end-to-end *task-equivalent to full context* (56.0% vs. 52.0%, McNemar $p = 0.688$) at a modest realised discount—the actionable conclusion being *keep compression recoverable*. The contract is right; the lesson is that proxy certification and lossy aggression are the failure mode, recoverability the fix.

Reproducibility. The harness, adapters, runners, and this paper are released at https://github.com/dshakes/distil (see `benchmarks/PROVE.md` and `docs/PAPER_PLAN.md`).

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